Lifetime Asset Allocation in New Zealand: A Simulation and Optimization Approach

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1 Foreword and Research Context

This research is conducted in partnership with Consilium NZ LTD (referred to hereon as simply Consilium). Consilium is a New Zealand-based financial services company that provides a range of services to financial advisers and their clients. Their services include a range of tools and consulting services to support advisers in areas such as investment management, portfolio construction, risk management, and financial planning. Consilium also offers a range of investment products, including managed funds and portfolio solutions, that are designed to help advisers meet the needs of their clients. The goal of this research partnership is to examine and improve lifetime asset allocation strategies for individual investors, with a particular focus on retirees in the New Zealand context. This research aims to provide insights and recommendations that can help Consilium better serve their clients and support financial advisers in delivering exceptional financial advice.

While this research is conducted in partnership with Consilium, the views and opinions expressed in this paper are those of the author and do not necessarily reflect the official policy or position of Consilium. The author retains full academic independence in the conduct of this research, including the design, analysis, and reporting of findings. The author also notes that they have agreed with Consilium to retain full ownership of the intellectual property rights associated with this research, granting Consilium a non-exclusive license to use the research findings for their internal purposes and to share them with their clients and partners. The author notes that no confidential or proprietary information provided by Consilium has been used in the conduct of this research.

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2 Introduction

Managing a retiree's nest egg requires navigating two broad forms of financial risk. A well-established principle in investment research is that assets with higher expected returns tend to come with higher volatility (e.g. Campbell, 1996; Ghysels et al., 2005; Lundblad, 2007). Although many investors spend considerable energy selecting particular securities or funds, Ibbotson and Kaplan (2001) find that roughly 90% of the variation in fund returns is explained by the allocation to major asset classes such as equities, bonds, and cash. As a result, asset allocation is the primary driver of a portfolio's long-term behaviour. For the typical investor, whose retirement savings are usually held in managed funds that do not allow meaningful security selection, the most important strategic decision is choosing the overall exposure to these broad asset classes.

The central challenge for individuals approaching retirement is finding an allocation between safe and risky assets that provides the returns required for a comfortable lifestyle, while avoiding excessive exposure to loss. Achieving this balance is not straightforward. Once the allocation is considered as a function of risk appetite, savings capacity, spending needs, lifespan, and total accumulated wealth, the problem becomes complex. Many of these inputs differ significantly between individuals, and even small differences in behaviour or circumstance can lead to materially different outcomes. The mathematical problem of finding an optimal allocation through time is therefore highly sensitive to real-world considerations. This complexity has contributed to the emergence of various heuristics and rules of thumb that attempt to provide simple guidance in the absence of a fully individualised solution.

The significance of this problem extends beyond individual welfare. The global asset-backed pension market was valued at 68.9 trillion USD in 2024 (OECD, 2025). For many people, their retirement savings represent the largest financial asset they will ever manage apart from their home. Consequently, the question of how to allocate this wealth over the course of a lifetime is significant not only for individuals but also for governments. Even modest improvements in the efficiency of retirement strategies can reduce pressure on public pension systems. In New Zealand, for example, every resident aged 65 or older receives New Zealand Superannuation regardless of income or assets. The programme is funded through general taxation. Due to an ageing population and ongoing increases in life expectancy, the long-term fiscal sustainability of this system is becoming increasingly challenging. A recent Treasury report (The Treasury, 2025) estimates that spending on New Zealand Superannuation and healthcare could account for nearly half of all government expenditure by 2065 if current policy settings remain unchanged. This has led to suggestions that New Zealanders will need to save more for retirement, for example through higher KiwiSaver contributions, and that adjustments to eligibility age or means testing may be required. There is already a broad consensus that the public is under-saving for retirement (e.g. Dexter, 2025; Gilbert, 2025; Luxon, 2025; Palmer, 2025).

A critical requirement for the success of such policies is ensuring that individuals make appropriate decisions about their lifetime asset allocation. At present, the New Zealand government provides little guidance on how individuals should allocate their retirement savings. The Financial Markets Authority requires that default KiwiSaver funds adopt a Balanced risk profile (Financial Markets Authority, 2023), defined as holding between 35 percent and 63 percent growth assets (Financial Markets Authority, 2024). This replaced the previous Conservative default in 2021 (Retirement Commission Te Ara Ahunga Ora, n.d.), after it became clear that many KiwiSaver members did not switch funds from the default fund and that a conservative allocation was poorly aligned with the needs of long-term investors (Ministry of Business, Innovation and Employment and The Treasury, 2019). Although a Balanced fund is a reasonable one-size-fits-most solution, it is likely too conservative for many younger investors who have long investment horizons and the capacity to tolerate short-term volatility.

This paper has three objectives. First, we review the current literature on lifetime asset allocation, with particular attention to the glide-path strategies that dominate both academic discussions and industry practice.

We also examine the different risk measures and optimisation frameworks that researchers have used to evaluate such strategies. Second, we analyse the spending needs and risk concerns of typical retirees, with a particular focus on the New Zealand context. Understanding these patterns is essential for designing realistic simulations of retirement behaviour and for evaluating the amount of risk that individuals can tolerate. Finally, we develop and test a flexible simulation and optimisation framework for lifetime asset allocation. Building on the work of Mäkinen and Toivanen (2024), who propose a Monte Carlo portfolio optimisation framework, we adapt and extend their method to address the specific challenges that arise in lifetime allocation problems. We also modify their risk measure so that it reflects the practical concerns of real-world investors.

Our aim is to show that this framework can generate stable and robust strategies, while remaining flexible enough to incorporate a variety of risk measures, asset return models, and spending assumptions. This should allow us to test the validity of existing glide-path recommendations and to provide a foundation for further research in this area of personal finance.

2.1 The Status Quo on Lifetime Asset Allocation

In order to understand the rationale for any allocation strategy, it is important to have a basic understanding of the financial life-cycle of the typical investor. Roughly speaking, an investor's life can be divided into two main phases: the accumulation phase and the decumulation phase. During the accumulation phase, which typically spans from early adulthood to retirement age, the investor focuses on building wealth through regular contributions to savings and investment accounts. During this period, the investor increases the wealth in their portfolio through a combination of contributions and investment returns. The decumulation phase begins at retirement and continues until the end of the investor's life. During this phase, the investor withdraws funds from their portfolio to cover living expenses and other financial needs. The goal during this period is to manage withdrawals and the investment's returns in a way that ensures the portfolio lasts throughout retirement. This two-phase framework also leads to a corresponding life-cycle wealth trajectory, in which wealth accumulates during the working years, peaks at retirement, and then might decline during retirement as funds are withdrawn. Although it is often assumed that this wealth trajectory is hump-shaped, with wealth declining steadily during retirement, to what extent real-world investor's wealth actually declines during retirement for many investors (or even whether ivestors wealth typically declines at all) is an open question that we will return to in a later section.

Another important concept in this discussion is human capital, which refers to the present value of an individual's future earnings potential. During the accumulation phase, human capital is typically high, as the investor has many working years ahead. As the investor approaches retirement, human capital declines, and at retirement, it effectively becomes zero. This shift in human capital has important implications for asset allocation decisions, as it affects the investor's overall risk profile and capacity to absorb investment losses.

The sheer ammount of complexity in finding optimal asset allocation strategies through the life-cycle is daunting. The optimal strategy depends on a wide range of factors, and likely because of this because this complexity, a large number of rules of thumb and financial products has emerged to attempt to guide investors and advisors. Most of these rules recommend a "declining-risk" strategy, in which exposure to equities decreases with age. This idea has been influential for decades and was even considered as part of potential reforms to the United States Social Security system under the George W. Bush administration (R. Shiller, 2005; R. J. Shiller, 2006). (The fact that it was considered for such a large and important public pension system is also a testament to the appeal to governments and policymakers of being able to improve retirement outcomes through better asset allocation strategies.)

A well-known example of this type of recommendation is the various "age in bonds" rules, which suggests holding a percentage of the portfolio in bonds equal or proportional to the investor's age. The investors' life-cycle allocation then would take the form of a lineary-decreasing function, with allocation to risk starting high and decreasing over time, hence "declining-risk" or "declining-equity" strategy. This general approach appears throughout investment textbooks (e.g. Bengen, 1996; Bodie et al., 2023) and in the popular financial advice literature (Choi, 2022). Dolvin et al. (2010) review U.S. target-date funds and find that the majority follow a some type of declining-risk structure that often resembles a 120 minus age equity allocation rule. In New Zealand, SuperLife's Age Steps product follows a similar pattern, reducing equity exposure from 99% at age step 18 to 10% at age step 80 (see Figure 2.1).

The intuitive logic behind declining-risk strategies is that older investors have less time to recover from losses, and that their spending needs become more immediate. In this view, reducing exposure to risky assets as retirement approaches is considered prudent. However, continuing to reduce risk exposure throughout retirement may not be optimal, and the academic literature support for this approach is mixed at best, and largely

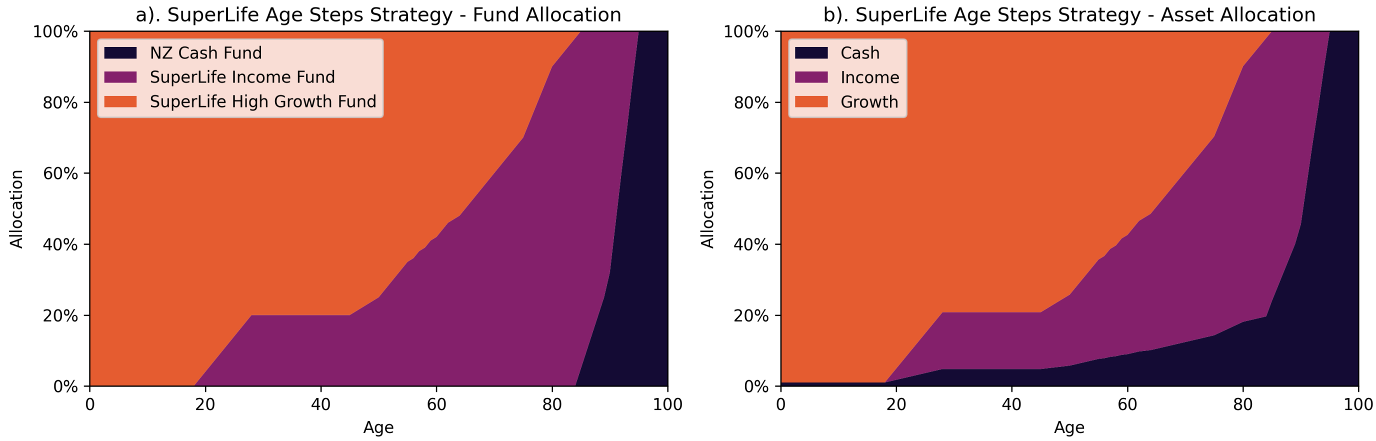

Figure 1: SuperLife's Age Steps target strategy as per their Statement of Investment Policy and Objectives (SIPO), showing the allocation to different funds (a) and the resulting allocation to different asset classes (b). Source: SuperLife (2026).

indicates that constant or even increasing-risk strategies may be more effective for most retirees undergoing decumulation.

The first mathematically rigorous work on this by Merton (1969) showed that when maximising consumption utility over a lifetime, the optimal asset allocation is constant through time and does not depend on age. However, this conclusion is based on the assumption of an investor who is withdrawing a fixed proportion of wealth each period (rather than a fixed dollar amount) and that asset returns are independent and identically distributed (IID) from year to year. Even so, modelling retirements suing empirical data, R. Shiller (2005) and R. J. Shiller (2006) found that declining-risk strategies may be inferior to a constant 100 percent equity allocation for retirees making fixed-dollar withdrawals. Moreover, Estrada (2014, 2015) find that declining-risk glide-paths tend to underperform both constant-risk and increasing-risk strategies across a wide range of risk measures, asset classes, and investor types. Using bootstrapped historical return data, they show that declining-risk strategies produce lower terminal wealth while providing no meaningful improvement in downside risk.

Although this paper will be principally concerned with the decumulation phase, it is worth noting that the evidence for declining-risk strategies during the accumulation phase is also mixed. The most robust theoretical support for declining-risk strategies during the accumulation phase comes from Bodie et al. (1992) and Cocco et al. (2005), who show that when human capital is taken into account, younger investors should hold a higher proportion of equities. In their models, human capital is treated as a bond-like asset because it provides a steady stream of income that is less volatile than equity returns. As investors age and their human capital declines, their overall risk exposure increases, which justifies a reduction in equity holdings. In this model, they essentially argue that the declining-risk strategy is appropriate during the accumulation phase, insofar as it offsets the changing risk profile due to declining human capital.

However, work done on emprical data suggests that even during the accumulation phase, declining-risk strategies may not be optimal. Anarkulova et al. (2025) find that for typical young investors with low wealth and savings rates, a constant 100 percent equity allocation provides similar downside protection to a declining-risk strategy while delivering substantially higher expected returns. This is an extremely rigorous and comprehensive study which we will be returning to multiple times throughout this paper. Using block-bootstrap resampled data from both U.S. and international markets, they find that optimised glide-paths still underperform portfolios invested entirely in equities, except in the very earliest years of retirement. They argue that fixed-income assets offer very limited benefit once time-dependent return dynamics are incorporated, and that earlier studies tended to favour bonds because they assumed returns are independent and identically distributed from year to year. In their optimal glide-paths, bonds appear only modestly and mostly at the start of retirement, with allocations typically in the range of 30 to 40 percent or less. This produces a U shaped risk trajectory in which risky allocations dip at retirement and climb thereafter.

This U-shaped risk trajectory is a common finding in the literature on optimal glide-paths, and it is often interpreted as evidence that some reduction in risk exposure at retirement is optimal. It appears again, more strongly, in work by Estrada (2015), who find that optimal glide-paths often involve a significant reduction in equity exposure at retirement, followed by a gradual increase in equity exposure later in retirement. This pattern is consistent with the idea that retirees are most vulnerable to sequence-of-returns risk at the start of retirement, when they have high accumulated wealth and low human capital. By holding some bonds at this point, retirees can reduce the likelihood of depleting their portfolio too quickly. However, as withdrawals reduce total wealth and the investor is further along in retirement, the impact of a run of bad returns becomes less severe (in real-dollar terms), and therefore a higher equity allocation becomes more acceptable in fixed-spending scenarios.

So while declining-risk strategies are widely recommended and implemented in practice for retirees, the academic and mathematic literature does not strongly support their use, particularly during the decumulation phase. Instead, evidence suggests that constant or even increasing-risk strategies may provide better outcomes for retirees. While decreasing risk exposure may be appropriate during the final years of accumulation due the investors' exposure to sequence-of-returns risk at this point, the evidence suggests that once retirees enter the decumulation phase, they may be better off maintaining or even increasing their exposure to risky assets.

The benefits of such strategies as implemented by products like SuperLife's Age Steps are likely to be limited, and may even be conterproductive for many retirees. The evidence suggests that the common-knowledge around lifetime asset allocation may be misguided, and that there is an opportunity for financial advisers to differentiate themselves by offering more sophisticated and evidence-based asset allocation strategies. However, it is important to note that the conclusions from the literature on this topic are not entirely consistent, and that the optimal strategy may depend on a variety of factors, including the specific risk measure used, the modelling of asset returns, and the treatment of investor behaviour with regard to spending and withdrawals. Additionally, many of the models used in the literature appear limited when considering real-world investor behaviour and market dynamics. Where conclusions from the literature diverge, it often stems from differences in the underlying assumptions and methodologies used in the analyses. Particularly as it relates to the choice of risk measure, the modelling of asset returns, and the treatment of investor behaviour with regard to spending and withdrawals.

Our goal will be to study and model investor concerns and behaviour in a way that is more realistic and relevant to the New Zealand context. We will then use this understanding to inform the design of a flexible simulation and optimisation framework for lifetime asset allocation, which can be used to test the validity of existing glide-path recommendations and to explore alternative strategies that may better meet the needs of New Zealand retirees.

Implications for Consilium NZ LTD

While Cosilium does not currently offer a glide-path product, nor do they explicitly recommend a declining-risk strategy, the findings from the existing literature have important implications for their business. It suggests that the current common-knowledge around lifetime asset allocation may be misguided, and that there is an opportunity for Consilium to differentiate themselves by offering more sophisticated and evidence-based asset allocation strategies. In particular, there may be an opportunity to challenge the assumptions of many of their advisors around need for declining-risk strategies.

Consilium has typically based much of its modelling of portfolio performance on IID return assumptions, which may not accurately reflect real-world market dynamics. By investigating asset allocation strategies that account for more realistic return dynamics, Consilium can provide more robust and effective advice to their clients.

3 Retirement in New Zealand

Because of the limited literature on the concerns New Zealanders in retirement specifically, we will also draw on international research to inform our understanding of the concerns of retirees. In doing so, it is important to outline the key economic and institutional features of the New Zealand retirement system, and to compare these to the systems in other countries that we will be drawing on for insights. New Zealand has a unique retirement system that is based on a combination of a universal public pension (New Zealand Superannuation) and voluntary private savings (KiwiSaver). The public pension provides a basic level of income to all retirees, regardless of their work history or savings, and is adjusted annually for inflation. KiwiSaver, on the other hand, is a voluntary savings scheme that allows individuals to contribute a portion of their income to a retirement savings account, with contributions from employers and the government. This system provides a safety net for all retirees while also encouraging personal savings and investment (Ministry of Social Development, 2022; Te Ara Ahunga Ora: Retirement Commission, 2021)

Given that Kiwisaver is relatively new, having been introduced in 2007, much of the current concerns of retirees in New Zealand are likely to shift as more people enter retirement with significant KiwiSaver balances. Currently, 40 percent of New Zealanders aged 65 rely entirely on New Zealand Superannuation for their retirement income, while 20 percent only have only a small amount of savings in addition to New Zealand Superannuation (Retirement Commission Te Ara Ahunga Ora, 2022). As more people enter retirement with significant KiwiSaver balances, this demography is likely to shift. To this end, the government is already taking steps to increase KiwiSaver contributions, which will further increase the amount of wealth that future retirees have in their KiwiSaver accounts (e.g. Luxon, 2025). This will likely lead to a shift in the concerns of retirees, as they will have more wealth to manage and will need to make more complex decisions about how to allocate their savings and how to draw down their wealth in retirement.

Because the literature on New Zealand retirees specifically is limited, we will also draw on international research to inform our understanding of the concerns of retirees. However, it is important to note that the institutional context of retirement in New Zealand is different from that of other countries, and therefore the concerns of retirees in New Zealand may differ from those in other countries. However, likely larger future role of KiwiSaver is beneficial for our research purposes because it will cause the New Zealand system to more closely resemble that of Australia.

Australia has a retirement system based on a combination of a means-tested public pension and a mandatory private savings scheme (superannuation). However, the Australian system goes significantly further than New Zealand in encouraging private savings, with a mandatory contribution rate of 12 percent of income, compared to New Zealand's voluntary contribution system, and with a significantly higher contribution rate for those who do choose to contribute (Australian Taxation Office, n.d.; Swoboda, 2014).

3.1 Spending By Age in New Zealand

By far the best source of data on spending patterns in retirement in New Zealand is the Stats NZ Household Economic Survey (HES). Two reports have been published using this data that are particularly relevant for our purposes. The first is a report by Le and Richardson (2023) that provides a detailed analysis of expenditure patterns by age in New Zealand. The second is a more recent report by the Retirement Income Interest Group (RIIG)(Retirement Income Interest Group, 2024) that focuses specifically on spending patterns through retirement and their implications for retirement planning and drawdown strategies.

Le and Richardson (2023) provide a detailed analysis of this phenomenon using New Zealand data. Using Data from the Household Economics Survey (HES) for the year ended 30 June 2019, they find that expenditure drops around retirement. Furthermore, consumption continues to decline throughout retirement. The mean total expenditure is $64,700 for households aged 65-74 and the mean for households aged 75-84 and aged 85+ are 28% and 44% lower respectively. Looking at the composition of expenditure, with the exception of rent and household content spending (furnishings etc.), all categories of expenditure decline with age. The largest declines between the younger and older retirees are in mortgage repayments, air travel, and recreation. Unusally compared to other countries, expenditure on healthcare also declines with age in New Zealand, which is likely due to the fact that healthcare is largely publicly funded in New Zealand, and therefore does not require out-of-pocket expenditure for most retirees. This is an important consideration for our modelling of retiree behaviour and concerns, as it suggests that healthcare costs are unlikely to be a major concern for most New Zealand retirees. Overall, Le and Richardson (2023) find that older retirees spend significantly less than younger retirees, particularly in descretionary categories. These findings are consistent with a more recent study based on more recent HES data (Retirement Income Interest Group, 2024) who find that, on average, retirement spending reduces by around 2% a year in real-dollar terms. That is, retirees' spending growth is below general price inflation by that amount for each year after age 65 or full retirement.

What this data cannot tell us, however, is why this decline in spending occurs. There are essentially two possibilities. First, it could be that retirees are voluntarily reducing their spending because they become less active and therefore have less need for goods and services. The second possibility is that retirees are involuntarily reducing their spending because they become financially constrained. Without data that tracks individual retirees over time, it is difficult to disentangle these two effects. The HES data is cross-sectional, so it cannot track the same individuals as they age. Therefore, it is possible that the observed decline in spending with age is due to a combination of both voluntary reductions in spending and involuntary reductions due to financial constraints.

This decline in spending with age is an important consideration for our modelling of retiree behaviour and concerns, as it suggests that retirees may have different spending needs and therefore risk tolerances at different stages of retirement. Furthermore, understanding why consumption falls - whether due to voluntary reductions in spending or involuntary reductions due to financial constraints - is important for understanding the welfare implications of this phenomenon. In other words, if the observed decline in spending at retirement a sign of a decline in welfare, then the optimal asset allocation strategy one that allows retirees to maintain their pre-retirement spending levels. On the other hand, if the observed decline in spending at retirement is due to voluntary reductions in spending (arising perhaps from the simple need to consume less), then it may not be a sign of a decline in welfare, and therefore the optimal asset allocation strategy may not need to focus on maintaining pre-retirement spending levels.

This decline in spending is a well-studied phenomenon in the economics of retirement and is sometimes known as the "retirement consumption puzzle". This refers to the repeated empirical finding that expenditure often drops when people retire, despite standard life-cycle and permanent-income models predicting relatively smooth consumption over time. In the canonical framework, households accumulate during working life and then gradually decumulate in retirement while maintaining living standards (Ando & Modigliani, 1963). In practice, many households appear to reduce spending at retirement and draw down wealth more slowly than these models would imply. This combination, both retirees spending less and drawing down wealth more slowly, is puzzling because it suggests a more complex set of dynamics than traditional models of consumption smoothing would predict.

To what extent this phenomenon reflects a true decline in welfare, as opposed to changes in spending patterns or measurement issues, is an open question. Levels of consumption are often used as a proxy for welfare, but they may not fully capture the well-being of retirees, especially when higher consumption is compelled by medical needs or lower consumption is voluntary due to lifestyle changes. For example, Barrett and Brzozowski (2010) find that involuntary reductions in spending play a role. They show that households that retire unexpectedly, especially after adverse health shocks, can experience sharper spending adjustments than households that retire as planned. This suggests that some but not all of the observed decline in expenditure at retirement may be due to financial constraints rather than voluntary changes in spending patterns. This helps explain why average effects can mask substantial cross-household dispersion in retirement outcomes. However, some of the observed decline in expenditure may be due to retirees voluntarily reducing their spending because they become less active and therefore have less need for goods and services. For example, retirees may spend less on commuting, work clothing, and purchased meals as they transition out of the workforce (Aguiar & Hurst, 2008; Battistin et al., 2009). Additionally, if retirees substitute market purchases with home production, observed expenditure may fall while underlying consumption remains closer to smooth-path predictions (Becker, 1965; Hurd & Rohwedder, 2003).

Cross-country comparisons reinforce the importance of institutional context. Banks et al. (2016) show that spending profiles at older ages differ materially between the US and UK, with retirees in the US having far higher consumption in late-life compared to their UK counterparts. Retirees in the UK reduced their real consumption by 2.2% per year compared to 1.4% per year in the US. The researchers find that differences in healthcare financing and pension design are key drivers of these divergent patterns, with US retirees facing higher out-of-pocket healthcare costs, which thus drives higher spending in this category, especially at older ages when healthcare needs typically increase. In contrast, UK retirees have more comprehensive public healthcare coverage, which reduces out-of-pocket costs and therefore leads to lower healthcare spending in retirement. This results in a more pronounced decline in overall expenditure for UK retirees compared to US retirees, however, when healthcare spending is excluded, the decline in expenditure at retirement is more similar between the two countries.

Declines in spending have also been observed in Australia. A review by the Grattan Institute (2019) finds that real spending declines by between 0.5% and 1% per year after retirement. Unlike the HES data for New Zealand, the Grattan Institute review examines longitudinal data, which allows them to track the same individuals over time. They find that the decline in spending is consistent between cross-sectional and longitudinal data, which suggests that the observed decline in spending with age is not solely due to differences between cohorts of retirees, but also reflects underlying changes in spending patterns as individuals age and is reflective of a less consumptive lifestyle.

Additionally, the Grattan Institute review also finds spending in Australia does not typically reduce in retirement due to financial constraints. Less than half of the retirees in their study were drawing down on their wealth at all and more than 40% of retirees were actually increasing their wealth during retirement. In fact, retirees were found to generally feel more financially comfortable than working-age people. As noted by Retirement Income Interest Group (2024), this suggests that the observed decline in spending at retirement is not primarily due to financial constraints, but rather reflects a combination of voluntary reductions in spending and changes in lifestyle and needs as individuals age.

3.2 Bequests and Precautionary Savings

An aspect of the "retirement consumption puzzle" that has also been studied is why many retirees do not draw down significantly on their wealth during retirement, and in some cases even accumulate more wealth during retirement. This is a well-established phenomenon, having been observed in multiple countries and datasets. For example, King (1982) observe it in Canada, and the aforementioned Grattan Institute review finds it in Australia, and it has been extensively studied within the US context (e.g. De Nardi et al., 2010; Hildebrand, 2001; Love et al., 2009).

An early explanation for this phenomenon (proposed by Hurd (1987)) is that the observed lack of decumulation is due to retirees' desire to leave bequests to their heirs. In this view, retirees are not drawing down on their wealth because they want to preserve it for their children or other beneficiaries. However, survey data and later studies have casted doubt on this explaination, and have offered alternatives. According to a survey of Australian and Dutch retirees by Alonso-García et al. (2022), bequeathment is not a significant concern for most retirees in either country, outside of leaving a request to their spouse. Instead, retirees cite maintaining their financial independence and having a good quality of life as their primary concerns. They conclude that the observed lack of decumulation might be due to the desire to hold onto precautionary savings for health and long-term care expenditures.

This so-called "precautionary" motive for saving is a well-supported theory in the literature. Modelling the risk that US retirees face from uncertain health and long-term care costs, De Nardi et al. (2010) find that the risk of large, unexpected health and long-term care costs explain a significant portion of the observed lack of decumulation. As we've already noted, Banks et al. (2016) find that healthcare costs are a significant driver of spending for US retirees, especially compared to UK retirees who have more comprehensive public healthcare coverage. As the report by Retirement Income Interest Group (2024) notes, New Zealand retirees face a semi-privatised healthcare system, in which most medical treatments are publicly funded, but there are still significant out-of-pocket costs for some services, especially some long-term care services (such as those for assisted living and memory care (New Zealand Government, n.d.)). This suggests that the precautionary motive for saving may be less pronounced for New Zealand retirees compared to those in the US, but precautionary savings, especially for long-term care costs, are likely to still be a significant concern for many New Zealand retirees.

3.3 Spending by Household Composition

4 Modelling Perceptions of Risk and Utility

Traditional portfolio optimisation (arguably starting with the work proposed by Markowitz (1952)) typically uses variance of asset returns as the risk measure. This idea is based on the assumption that investors are risk-averse and that they prefer portfolios with lower variance for a given level of expected return. However, when we consider lifetime asset allocation, this type of measure is difficult to apply and interpret. For example, the variance of returns does not directly capture the risk of running out of money in retirement, it is instead the effect of this variance (in conjuction with consumption needs) that leads to the risk of ruin. Additionally, translating variance into something that is meaningful for real-world investors is difficult, and determining one's "risk tolerance" in terms of variance is not straightforward. For these reasons, using variance as the risk measure for optimising lifetime asset allocation strategies is likely to lead to suboptimal and potentially misleading recommendations.

This idea does persist, a version of it makes an appearance in Mäkinen and Toivanen (2024), who demonstrates their method using variance and semi-variance of terminal wealth as the risk measure for optimising their control-matrix asset allocation strategy. While probably intended as a proof-of-concept, this choice of risk measure leads to some odd results, where the optimiser ends up truncating the distribution of terminal wealth in a way that is not necessarily aligned with the concerns of real-world investors. A more in-depth discussion of their paper, and a replication of their results, is provided in the appendix.

Ultimately, it is important that whatever risk measure we choose, it should make sense from a behavioural perspective and, ideally, should be something that investors can easily understand and relate to. In constructing this risk measure, actually understanding the concerns of real-world investors is crucial for selecting an appropriate risk measure. These concerns are also inevitably tied into how retirees also choose or need to spend their wealth during retirement, as the risk of not being able to meet spending needs is likely to be a primary concern for most retirees. The rest of this section will be first dedicated to a review of what literature exists on the spending patterns and financial concerns of retirees, with a particular focus on the New

Zealand context, and then we will discuss how we can use this understanding to inform our choice of risk measure for optimising lifetime asset allocation strategies, with a particular focus on developing one that is both interpretable, communicable, and relevant to typical New Zealanders.

An essential component of optimising asset allocation strategies is the choice of risk measure. The risk measure serves as the objective function that we seek to minimise (or maximise) when searching for the optimal strategy. The choice of risk measure can significantly influence the resulting optimal strategy, as different risk measures capture different aspects of risk and may lead to different trade-offs between return and risk.

Our risk measure should be both interpretable and relevant to the concerns of real-world investors. It should be both easy to understand and directly related to the outcomes that investors care about. Additionally, depending on the optimistation method we choose, the risk measure may also need to be differentiable with respect to the parameters of the asset allocation strategy, which would allow us to use gradient-based optimisation techniques.

5 Methods for Simulating Lifetime Wealth Trajectories

A key component of studying lifetime asset allocation is the ability to simulate realistic wealth trajectories for individual investors. This involves modelling both the accumulation and decumulation phases, taking into account factors such as contributions, withdrawals, investment returns, and spending needs. Our approach builds on a Monte Carlo simulation framework, which allows us to generate a wide range of possible outcomes based on different asset allocation strategies and market conditions. Importantly, the smulation approach we propose is agnostic to the source of asset returns, meaning that we can use historical data, bootstrapped returns, or simulated returns from a variety of models. This flexibility allows us to test the robustness of different asset allocation strategies across a range of assumptions. Additionally, the framework also allows for the incorporation of deposits and withdrawals of different amounts at different times, which is a key feature of real-world investor behaviour.

Unlike the approach proposed by Merton (1969), ours is a discrete-time simulation that models the investor's wealth at regular intervals. This allows us to capture savings and spending behaviour in fixed-dollar amounts, which is more realistic for most retirees.

Each step of the model takes the following form:

$$W_{t+1} = g_{t+1} W_t + \pi_{t+1} \tag{1}$$

Where W_t is the wealth at time t, g_{t+1} is the growth factor on the portfolio between time t and $t+1$, and π_{t+1} is the net contribution (or withdrawal if negative) made at time $t+1$. This can represent regular contributions during the accumulation phase or withdrawals to cover spending needs during the decumulation phase. The model can be easily adapted to include varying contribution and withdrawal amounts over time, allowing for a more nuanced representation of investor behaviour, including simulating variable or stochastic spending patterns.

The growth factor g_{t+1} is determined by the asset allocation strategy being tested, which specifies the proportion of wealth invested in different asset classes. Specifically, we define asset allocation as a vector $\mathbf{a}_t$ at time t, where each element $a_{i,t}$ represents the proportion of wealth invested in asset class i. The portfolio growth factor is then calculated as:

$$g_{t+1} = 1 + \mathbf{a}_t^\top \mathbf{r}_{t+1} \tag{2}$$

Where $\mathbf{r}_{t+1}$ is a vector of returns for each asset class between time t and $t+1$.

In a Monte Carlo simulation, we generate a large number of possible return paths. This means that for each time step, we have N different return vectors for our k asset classes. For each path, we simulate the investor's wealth trajectory over time using the equations above. This means that equation 1 has the matrix form:

$$\mathbf{w}_{t+1} = \mathbf{w}_t \odot \mathbf{g}_{t+1} + \boldsymbol{\pi}_{t+1} \tag{3}$$

Where $\mathbf{w}_t$, $\mathbf{g}_{t+1}$, and $\boldsymbol{\pi}_{t+1}$ are vectors of length N representing the wealth, portfolio returns, and contributions or withdrawals for each of the N simulated paths at time t, and $\odot$ denotes element-wise (Hadamard) multiplication.

The growth factors $\mathbf{g}_{t+1}$ are determined by the asset allocation strategy and a $N \times k$ matrix of returns $\mathbf{R}_{t+1}$ for the k asset classes across the N simulated paths. The asset allocation strategy can be defined in various ways, the first of which is as a function of time (or age). In this case, we define the asset allocation at time t as $\mathbf{a}_t = f(t; \boldsymbol{\theta})$, where f is a function parameterised by a set of parameters $\boldsymbol{\theta}$.

This yeilds the portfolio growth factors as:

$$\mathbf{g}_{t+1} = 1 + \mathbf{R}_{t+1} \cdot f(t+1; \boldsymbol{\theta}) \tag{4}$$

In this form, the model can accommodate a wide range of asset allocation strategies based on the investor's age. This is similar to the approach that Anarkulova et al. (2025) use, and our first objective will be to replicate their findings for the optimal glide-paths, and to examine how these findings change when we modify the risk measures and spending assumptions to better reflect real-world investor behaviour, particularly in the New Zealand context.

Implications for Consilium and an Aside on Using Matrix Operations and Computational Efficiency

The effort taken here to express the model in matrix form is not only for academic rigor, but it also allows us to leverage vectorised operations through highly optimised linear algebra libraries (e.g. `numpy` (Harris et al., 2020)) which can significantly speed up the simulation process, especially when dealing with a large number of paths and time steps. A mistake made in previous work by Consilium when it has considered using this type of Monte-Carlo approach to simulating investor outcomes is that they have constructed very poor computational implementations that rely heavily on for-loops and other non-vectorised operations, which can be extremely slow and inefficient. By contrast, using matrix operations allows us to perform calculations on entire arrays of data at once, leading to many orders of magnitude improvement in speed. This work alone, if applied to Consilium's existing offerings would represent a significant improvement in the service it provides to its clients.

The use of matrix operations will also make it easier when it comes to optimising the asset allocation strategy, as we can leverage automatic differentiation tools (e.g. `tensorflow` (Martín Abadi et al., 2015), `pytorch` (Paszke et al., 2019)) to compute gradients of the risk measure with respect to the parameters of the asset allocation strategy.

5.1 Modelling Dynamic Investor Behaviour

The model described above is flexible enough to accommodate a wide range of asset allocation strategies, but it also allows for the modelling of dynamic investor behaviour in terms of contributions and withdrawals. This is an important feature, as it allows us to simulate more realistic spending patterns that can vary over time and in response to changes in the investor's wealth.

5.1.1 Wealth-Responsive Spending Policy

Rather than having a fixed spending pattern, we can also model the investor's spending as a function of their current wealth level. In some cases, retirees may have a fixed spending floor that they need to meet each year, regardless of the performance of their portfolio. In other cases, retirees may adjust their spending based on the performance of their portfolio, for example by reducing discretionary spending during periods of poor returns. To model this behaviour, we can define the net contribution (or withdrawal) at time t as a function of wealth: $\pi_t = g(W_t; \boldsymbol{\phi})$, where g is a function parameterised by a set of parameters $\boldsymbol{\phi}$. This allows us to simulate more realistic spending patterns that can vary over time and in response to changes in the investor's wealth. For example, we could model a spending policy that has a fixed floor and a variable discretionary component that adjusts based on the investor's current wealth level. This could be represented as:

$$\pi_t = -\max(S_{\text{floor}}, \alpha W_t) \tag{5}$$

Where S_{floor} is the fixed spending floor, and α is a parameter that determines how much of the investor's wealth is allocated to discretionary spending.

Additionally, we can prevent the investor from withdrawing more than their current wealth by defining the net contribution as:

$$\pi_t = -\min(g_{t+1} W_t, \max(S_{\text{floor}}, \alpha W_t)) \tag{6}$$

Equally, we could model a spending policy that has a random component, for example by adding a stochastic term to the spending function modelling rare, but significant, unexpected expenses that retirees may face, such as healthcare costs or home repairs.

5.1.2 Wealth-Responsive Allocation Policy

Like with the spending policy, we can also model the asset allocation strategy as a function of both time and wealth. In the age-based glide-path framework, the hypothetical investor's asset allocation strategy is only responsive to time (i.e. age), and does not take into account the investor's current wealth level. In practice, an investor's risk tolerance and spending needs may vary significantly depending on their accumulated wealth at any given time. For example, an investor with a large portfolio may be able to tolerate more risk than an investor with a smaller portfolio, even if they are the same age. To address this limitation, we will attempt to extend the framework using a method similar to that proposed by Mäkinen and Toivanen (2024), who introduce a control matrix that allows the asset allocation strategy to depend on both time and wealth.

In this framework, the asset allocation at time t and wealth W_t is given by $\mathbf{a}_t = C(t, W_t; \boldsymbol{\theta})$, where C is the control matrix parameterised by $\boldsymbol{\theta}$. This yeilds the portfolio growth factors as:

$$\mathbf{g}_{t+1} = 1 + \mathbf{R}_{t+1} \cdot C(t+1, W_t; \boldsymbol{\theta}) \tag{7}$$

The way that this control-matrix works is that it defines a grid over the time and wealth dimensions, with each cell in the grid corresponding to a specific asset allocation. During the simulation, the investor's current age and wealth level are used to determine which cell in the grid they fall into, and thus the corresponding allocation to the risky asset (the remainder of the portfolio is allocated to the safe asset).

This approach allows for a more nuanced asset allocation strategy that can adapt to the investor's changing circumstances over time. For example, an investor who has accumulated significant wealth may be able to take on more risk, while an investor with lower wealth may need to adopt a more conservative approach. However, it also exponentially increases the number of parameters that need to be optimised, which can make any optimisation process more challenging.

However, we can reduce the search-space somewhat by using less granular grid points for the control matrix and interpolating between them to determine the asset allocation for time steps and wealth levels that fall between the defined grid points. This allows us to maintain a reasonable level of flexibility in the asset allocation strategy while keeping the number of parameters manageable. Moreover, it is unlikely that investors' asset allocation policy will be meaningfully different for small increments in time or wealth, so using a coarser grid with interpolation is unlikely to significantly reduce the optimality of the resulting strategy, while making the optimisation process more tractable. This idea of interpolating between grid points was first proposed by Mäkinen and Toivanen (2024), who use bilinear interpolation to determine the asset allocation for wealth levels that fall between the defined wealth-grid points, however, we also propose that interpolation could be used for time steps that fall between the defined time-grid points as well, which would further reduce the number of parameters that need to be optimised.

To understand how this works on a more practical level, consider figure 2, which shows a simple example of a control matrix for a two-asset portfolio. Our control matrix has 6 time steps (0 to 5) and 5 wealth levels (0 to 4), so is a 6 by 5 matrix. Each cell in the matrix corresponds to a specific allocation to the risky asset, with the remainder allocated to the safe asset. For example, if an investor is at time step 2 and has a wealth level of 3, they would fall into the cell corresponding to an allocation of 50 percent to the risky asset and 50 percent to the safe asset. For timesteps or wealth levels that fall between the defined grid points, we can use interpolation method proposed by Mäkinen and Toivanen (2024) to determine the appropriate asset allocation. For example, if an investor is at time step 2.5 and has a wealth level of 3.5, we could use bilinear interpolation to calculate the asset allocation based on the four surrounding grid points.

For allocations with more than two asset classes, the control matrix can be extended to a higher-dimensional tensor, with each additional dimension representing an additional asset class. Mäkinen and Toivanen (2024) use bilinear interpolation to determine the asset allocation for wealth levels that fall between the defined grid points, ensuring a smooth transition between different allocation strategies.

5.2 Methods for Simulating Asset Returns

6 Methods for Optimising Asset Allocation Strategies

The key challenge in optimising the asset allocation strategy is that the the parameter space can be very large, even for relatively simple strategies. For example, if we use are searching for an optimal age-based glide-path with 2 asset classes, with allocations in 10 percent increments across a 60 year period, this results in 11 possible allocations at each of 60 time steps, resulting in a total of 11^{60} possible strategies. This is clearly an intractably large search space, and so we must use numerical optimisation techniques to find a good solution.

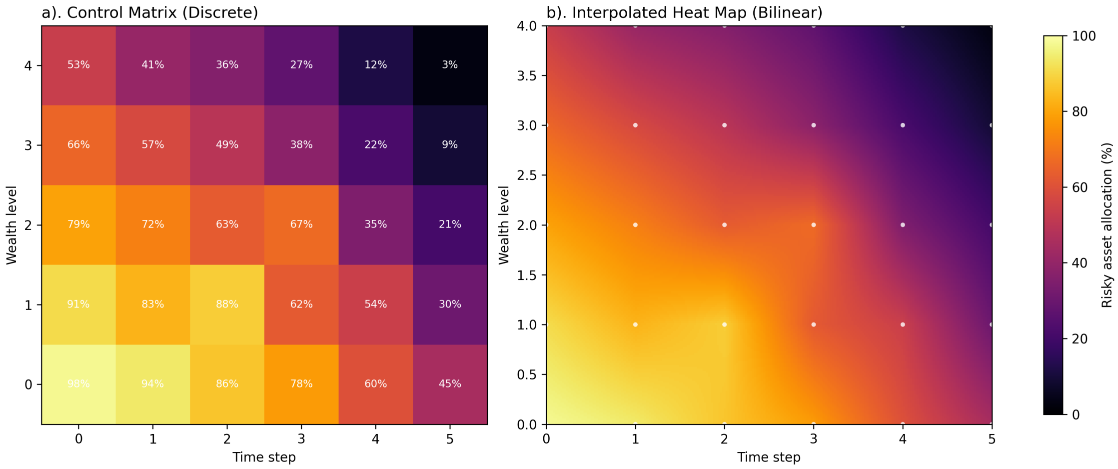

Figure 2: Illustrative example of a discrete control matrix (a) and bilinearly interpolated allocation surface (b).

Merton (1969) use a formulation where the investor maximises a utility function over their lifetime, based on the consumption at each time step. This approach has the advantage of being mathematically tractable, but it relies the investor consuming a fixed proportion of their wealth at each time step, which is not realistic for most retirees. Additionally, its closed-form solution assumed that asset returns are IID, which is the type of return dynamics that Anarkulova et al. (2025) argue leads to misleading conclusions about the optimality of declining-risk strategies, and so this approach is not suitable for our purposes.

Anarkulova et al. (2025) use a recursive approch where a grid-search is used to find the optimal allocation at each time step, working forwards in passes until convergence is reached. This approach will likely work for the age-based glide-path optimisation, but it is not clear how it would extend to the control-matrix approach, where the asset allocation depends on both time and wealth, as this increases the number of parameters by an order of magnitude again.

In their paper, Mäkinen and Toivanen (2024) optimised their control-matrix asset allocation strategy

7 Optimal Lifetime Asset Allocation Strategies for New Zealand Retirees

7.1 Optimal Age-Based Glide-Paths

7.2 Optimal Wealth-Responsive Allocation Strategies

8 Discussion and Implications

9 Conclusion

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A Replication and Discussion of Control Matrix Optimisation as Proposed by Mäkinen and Toivanen (2024)

Mäkinen and Toivanen (2024) formulate their monte-carlo optimization very similarly to the way we have described in the main body of this report. Like us, they use a control matrix to allow the asset allocation strategy to respond to both investor age and wealth. Their optimisation method is defined in terms of a cost function that is based on the terminal wealth of the investor, in their case the varaince or semi-variance of terminal wealth. They then efficiently compute the gradient of this cost function with respect to the parameters of the control matrix using adjoint methods, which allows them to use gradient-based optimization techniques (in this case, a quasi-Newton method with the BFGS update rule) to find the optimal control matrix.

Because this method is mathematically elegant and computationally efficient, we have attempted to replicate it here. The code for this replication can be found in the accompanying repository, and the results are shown in figure 3 for the mean-variance and mean-semi-variance cases.

In line with the setup described by Mäkinen and Toivanen (2024), the replication uses a Monte Carlo horizon of $T = 20$ years with time step $\Delta t = 0.25$, giving $N = 80$ time steps. The policy is represented on a time-wealth control grid with $M = 31$ wealth nodes, upper wealth bound $W_{\max} = 30$, and bilinear interpolation between nodes. In the notebook, wealth is clipped to $[0, W_{\max}]$ before interpolation. Optimisation is performed with L-BFGS-B (a quasi-Newton method) with box constraints.

For the one-risky-asset, no-leverage case used in figure 3, the full parameterisation is summarised in Table 1.

Table 1: Parameter setup for the one-risky-asset mean-variance replication.

Parameter	Value
Risk-free rate (r)	0.03
Risky-asset drift (μ_1)	0.0795
Risky-asset volatility (σ_1)	0.15
Contribution rate (π)	0.1
Initial wealth (w_0)	1
Minimum risky allocation ($p_{\min}$)	0
Maximum risky allocation ($p_{\max}$)	1
Number of simulation paths (K)	10^5
Risk-aversion weight (λ)	0.1

The results of the replication are broadly in line with figures produced by Mäkinen and Toivanen (2024) themselves. This replication highlights some important limitations in their method, however. Firstly, as a cost-funtion for a pension investor, the variance of terminal wealth is not a particularly good measure of risk, and its use here highlights some of its oddities. Looking at the histogram produced by the simulation, we can see that the distribution of terminal wealth is has a bi-modal distribution, which is the product of the fact that the allocation policy is switches to an extremely conservative allocation for high wealth levels, which results in a the normally long tail of the typically log-normal distribution of terminal wealth being truncated, and instead being condensed into a large number of outcomes at a single point in the distribution. This is not something that would be expected to be optimal for a real-world investor, but does make sense when the optimiser is “targeting” a certain variance of terminal wealth. This applies, albeit to a lesser extent, to the semi-variance case as well. While semi-variance, which only penalises outcomes below a the mean, would be expected to more closely align with investor preferences, it still results in a similar truncation of the distribution of terminal wealth, albeit at a higher wealth level than the mean-variance case. This is because higher portfolio returns result in a higher mean thus increasing the semi-variance penalty, causing the optimiser to, again, truncate the distribution of terminal wealth. This highlights the importance of choosing a risk measure that is both interpretable and relevant to real-world investors, as the choice of risk measure here is producing results that are not only counterintuitive, but also unlikely to be optimal for real-world investors.

Additionally, we can see from the results that the control matrix has very little control over the early stages of the simulation, as a large number of outcomes are condensed onto just a few points in the control matrix. This is because the wealth levels at the beginning of the simulation are much lower than the upper bound of the control matrix, which means that many of the simulated paths fall into the same wealth node in the control matrix, giving the optimization a limited ability to decern between different trajectories early on, which is precisely when intervention will have the greatest effect due to the compounding nature of asset returns. As a result, their method has too little control early on, which is arguably when it needs it the most. This also results in a large number of redundant parameters which sit at wealth levels that are never reached by the simulated paths (the upper-left portion of the matrix in figs 3). It is for this reason that we propose using a control matrix with an upper bound that is based on the distribution of maximum wealth outcomes in the simulation, which would ensure that the control matrix has sufficient resolution across the range of wealth levels that are actually reached by the simulated paths, while also reducing the number of redundant parameters and giving the optimization more ability to decern between different trajectories early on.

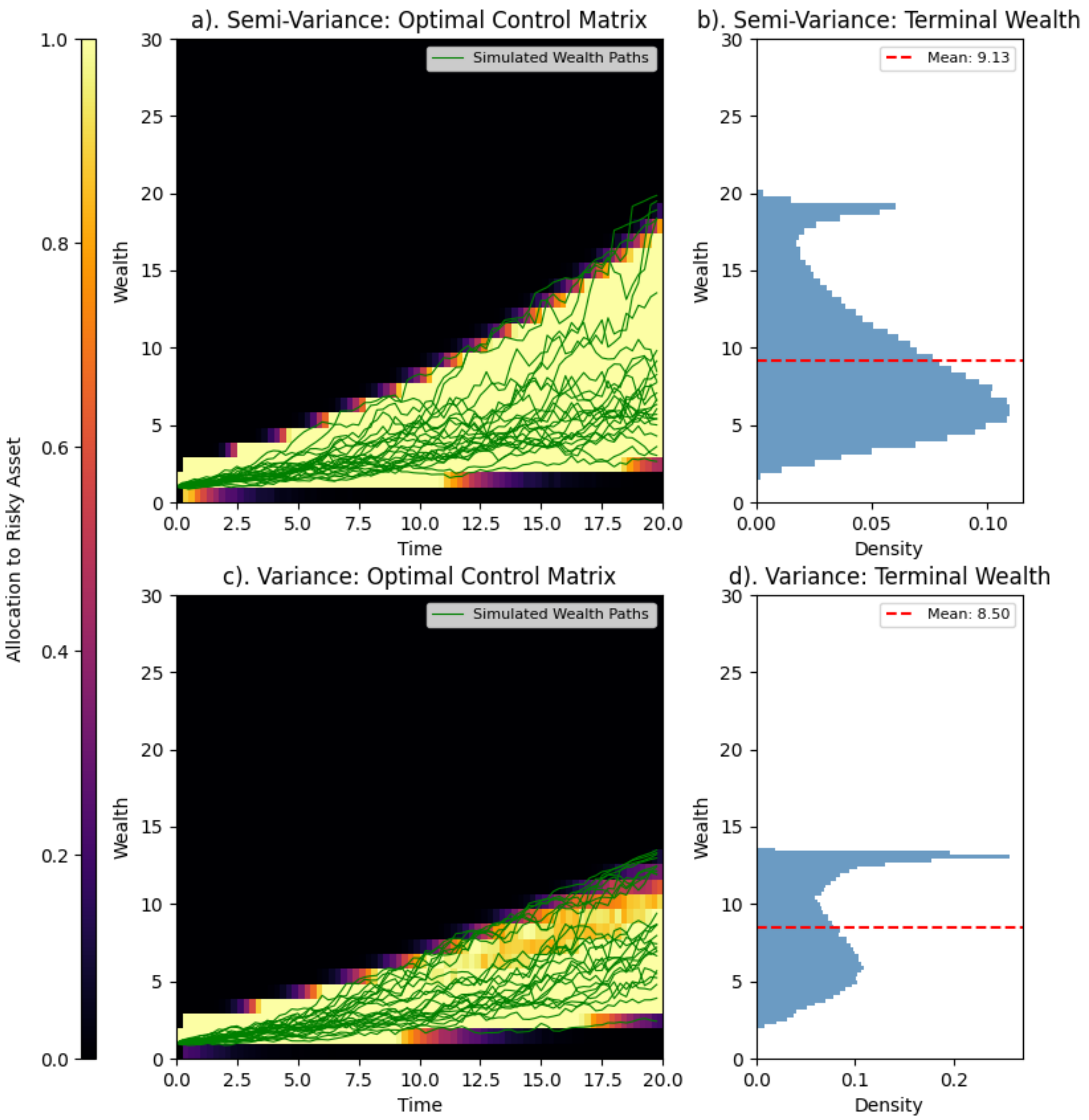

Figure 3: Replication of the control matrix optimization method proposed by Mäkinen and Toivanen (2024) for one risky asset and no leverage. The left column shows the optimal control matrix for the mean-semi-variance case (top) and the mean-variance case (bottom). The right column shows the corresponding histograms of terminal wealth for each risk measure respectively.